Koa Shen

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Cisco Hardware Hiring Team

Re: Summer 2027 Hardware Engineering Internship

Dear Cisco Hardware Hiring Team,

I am a mechanical engineering student at UC Santa Barbara, graduating in June 2028, and I am applying for a Summer 2027 hardware engineering internship at Cisco. I am drawn to the engineering behind dependable networking infrastructure, where hardware has to work as an integrated system and remain serviceable over time. My experience brings together mechanical design, simulation, embedded controls, fabrication, and hands-on validation, and I am eager to contribute that perspective while learning from Cisco's hardware teams.

At SLAC National Accelerator Laboratory, I built a Python/Drake digital twin for a tightly packaged beamline assembly with 30 degrees of freedom and 305 bodies. Millisecond collision checks exposed at least five physical pinch points and informed mechanical redesigns. I also developed 15 reusable motion-stage models, now being adopted across three LCLS assemblies, and redesigned mounts, alignment features, and detector strain relief. By pairing simulation with load checks and Solid Edge designs documented with GD&T, I learned to turn system-level risks into changes that can be built and inspected.

With UCSB Gaucho Racing, I led 15 teammates through safety-system design, manufacturing, and installation; our car passed FSAE technical inspection fully rules-compliant. I CNC-machined 33 suspension parts from 4130 steel and 7075-T6 aluminum, holding critical bearing fits within 0.0005 inch. In a personal actuator project, I designed a compact 15:1 reducer and wrote C++ firmware for stepper control and encoder feedback, then bench-validated position correction and missed-step detection. These projects taught me to work across mechanical and electrical interfaces and to verify behavior on real hardware.

I would bring Cisco a practical, systems-minded engineer who is comfortable moving between CAD, code, the machine shop, and test. I value clear documentation and designs that other engineers can assemble, inspect, and maintain. I would welcome the opportunity to contribute that mindset to hardware that keeps people and organizations connected.

Thank you for your consideration.

Sincerely,
Koa Shen